

Competency

- To solve the daily life problems on profit and loss by using fundamental rules of profit/loss and formulae.

Competency

- To solve the daily life problems on by using fundamental rules of commercial mathematics and formulae

Learning Outcomes

- To solve the daily life problems related to Commission, Tax and Dividend.

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define commission - To tell the formula of finding the commission amount - To define bonus - To tell the formula of finding the dividend - To define income tax
2.	Understanding (U)	- To calculate the commission amount - To calculate the bonus amount received by each employee
3.	Application (A)	- To find the commission or total sales - To find the income tax. - To solve the verbal problems related to dividend
4.	High Ability (HA)	- To compare the income taxes paid by an individual and the married couple. - To mathematize the daily life problems about the shares of some companies or banks or business solve them.

Required Teaching Materials/ Resources

Colourful chart-paper with definitions of buyer, seller, agent, commission and formulae, bills, income tax rates published by IRD etc

Pre-knowledge: cost price, selling price, profit and loss, discount and VAT

A. Bonus

Teaching Activities

1. Start the classroom discussion with the following questions:
 - (i) Have you heard about bonus?
 - (ii) Have anyone of your family members received bonus yet?
 - (iii) What is Net profit?
 - (iv) How do you define bonus?

2. Explain bonus as the extra amount of money that is distributed to the employees by the company for their goods performance from the certain percent of net profit of the fiscal year.
3. Focus on group work to let the students formulate the concepts through examples.
4. Ask the formula of finding the bonus amount , bonus amount received by each employee, bonus , explain the following formulae
 - (i) Bonus amount = Bonus % of net profit
 - (ii) Bonus amount received by each employee = $\frac{\text{Total bonus amount}}{\text{No. of employee}}$
 - (iii) Bonus percent = $\frac{\text{Bonus amount}}{\text{Net profit}} \times 100\%$
 - (vi) Annual income = annual salary + bonus

Solution of selected problems from Vedanta Excel in Mathematics

1. **A publication housed announced to distribute 10% bonus equally to its 20 employees from the net profit of Rs 18,36,000 at the end of a fiscal year, find the bonus received by each employee.**

Solution:

Here, net profit of the company = Rs 18,36,000, bonus percent = 10%, number of employees = 20

Now, bonus amount = 10% of Rs 18,36,000 = Rs 1,83,600

Bonus amount received by each employee = $\frac{\text{Rs } 1,83,600}{20} = \text{Rs } 9,180$

Hence, each employee received Rs 9,180.

2. **A garment factory announced 20% bonus to its 25 workers from the net profit at the end of last fiscal year. If every worker received Rs 18,500, how much was the profit of the factory?**

Solution:

Here, bonus percent = 20%, number of employees = 25, bonus amount received by each worker = Rs 18,500

Now, bonus amount = $25 \times \text{Rs } 18,500 = \text{Rs } 4,62,500$

Let net profit of the factory be Rs x

Then, bonus amount = bonus % of net profit

or, Rs 4,62,500 = 20% of x = 0.2x

$\therefore x = \text{Rs } 23,12,500$

Hence, the profit of the factory is Rs 23,12,500.

3. **A business company distributed bonus to its 24 employees from the net profit of Rs 16,48,000. If every employee received Rs 8,240, what was the bonus percent?**

Solution:

Here, net profit of the company = Rs 16,48,000, number of employees = 24, bonus amount received by each worker = Rs 8,240, bonus percent = ?

Bonus amount = $24 \times \text{Rs } 8,240 = \text{Rs } 1,97,760$

Now, bonus percent = $\frac{\text{Bonus amount}}{\text{Net profit}} \times 100\% = \frac{\text{Rs } 1,97,760}{\text{Rs } 16,48,000} \times 100\% = 12\%$

Hence, the required bonus percent was 12%.

4. **A garment factory made a net profit of Rs 48,00,000 in the last year. The management of the factory decided to distribute 18% bonus from the profit to its 25 employees.**
 - (i) Find the bonus amount received by each employee.
 - (ii) By what percent should the bonus be increased so that each employee can receive Rs 38,400?
 - (iii) What should be the profit of the company so that it can provide Rs 40,000 to each at 15% bonus?

Solution:

Here, net profit of the company = Rs 48,00,000, bonus percent = 18%
 number of employees = 25

(i) Bonus amount = 18% of Rs 48,00,000 = Rs 8,64,000

$$\text{Bonus amount received by each employee} = \frac{\text{Rs } 8,64,000}{25} = \text{Rs } 34,500$$

Hence, each employee received Rs 34,500.

(ii) Net profit of the company = Rs 48,00,000, number of employees = 25, bonus amount received by each employee = Rs 38,400, bonus percent = ?

$$\text{Bonus amount} = 25 \times \text{Rs } 38,400 = \text{Rs } 9,60,000$$

$$\text{Bonus percent} = \frac{\text{Bonus amount}}{\text{Net profit}} \times 100\% = \frac{\text{Rs } 9,60,000}{\text{Rs } 48,00,000} \times 100\% = 20\%$$

Hence, the bonus should be increased by $20\% - 18\% = 2\%$ so that each employee can receive Rs 38,400.

(iii) Bonus percent = 15%, number of employees = 25, bonus amount received by each worker = Rs 40,000

$$\text{Now, bonus amount} = 25 \times \text{Rs } 40,000 = \text{Rs } 10,00,000$$

Let net profit of the factory be Rs x

Then, bonus amount = bonus % of net profit

$$\text{or, Rs } 10,00,000 = 15\% \text{ of } x = 0.15x$$

$$\therefore x = \text{Rs } 66,66,666.67$$

Hence, the profit of the factory is Rs 66,66,666.67

5. **When a publication house increased its profit from 20% to 25%, the amount of profit increased to Rs 52,08,000. If the company decided to distribute 60% bonus to its 30 employees equally from the increased amount of profit, how much bonus will each employee receive?**

Solution:

Let the yearly income of the publication house be Rs x

Then, 25% profit of yearly income = Rs 52,08,000

$$\text{or, } 0.25x = \text{Rs } 52,08,000$$

$$\therefore x = \text{Rs } 2,08,32,000$$

Also, 20% profit of the yearly income = 20% of Rs 2,08,32,000 = Rs 41,66,400

Increased amount of profit = Rs 52,08,000 – Rs 41,66,400 = Rs 10,41,600

Now, bonus amount = 60% of Rs 10,41,600 = Rs 6,24,960

$$\therefore \text{Bonus amount received by each employee} = \frac{\text{Rs } 6,24,960}{30} = \text{Rs } 20,832$$

Hence, each employee received Rs 20,832.

Extra Questions

1. A noodle factory announced to distribute 10% bonus equally to its 45 employees from the net profit of Rs 34,65,000 at the end of a fiscal year, find the bonus received by each employee. [Ans:Rs 7700]
2. A garment factory announced 20% bonus to its 70 workers from the net profit at the end of last fiscal year. If every worker received Rs 15,000, how much was the profit of the factory? [Ans:Rs 52,50,000]
3. The management of a supermarket decided to distribute a bonus to its 180 employee from the net profit of Rs 3,00,00,000 at the end of a fiscal year. If every employee received Rs 25,000, what was the bonus percent? [Ans:15%]

B. Commission

Teaching Activities

1. With story/ related examples on commission, explain the words sales, agent and commission.
2. Define commission as the amount of money paid to the agent for performing the business service such as buying and selling goods, property (land, building, car etc.) or collection of money.
3. Under discussion, explain the following formulae
 - (i) Commission amount = commission % of total sales
 - (ii) Commission percent = $\frac{\text{Commission amount}}{\text{Total sales}} \times 100\%$
Monthly income = salary + commission

Solution of selected problems from Vedanta Excel in Mathematics

1. ***A real estate company gives 5% commission on selling a piece of land for Rs 10,00,000 and 7% commission for the additional amount of selling price above the fixed price. If the agent sold the land for Rs 12,99,000, how much commission did he/she receive from the company?***

Solution:

Here, the fixed selling price of the land = Rs 10,00,000

The selling price of the land = Rs 12,99,000

Now, the commission received by the agent = 5% of Rs 10,00,000 + 7% of (Rs 12,99,000 – Rs 10,00,000) = Rs 50,000 + Rs 20,930 = Rs 70,930

Hence, the agent received the commission of Rs 70,930.

2. ***The monthly salary of a sales person of a subway restaurant is Rs 21,600 and an additional incentive of 1.5% on the total monthly sale is provided as commission.***
 - (i) ***Calculate his/her total income in a month if he/she makes a total sale of Rs 5,80,000 in the month.***
 - (ii) ***What should be his/her total sale in the next month so that he/she can receive a total income of Rs 31,350 in the month?***

Solution:

Here, the monthly salary of a sales person = Rs 21,600

- (i) The total sales of the month = Rs 5,80,000 and commission percent = 1.5%

Now, the commission received by the sales person = 1.5% of Rs 5,80,000 = Rs 8,700

Hence, the income of the sales person in the month = salary + commission
= Rs 21,600 + Rs 8,700 = Rs 30,300

- (ii) The income of the next month = Rs 31,350

∴ Commission amount received in the next month = Rs 31,350 – Rs 21,600 = Rs 9,750

Let the total sales of the next month be Rs x.

Then, the commission amount = commission % of total sales

or, $\text{Rs } 9,750 = 1.5\% \text{ of } x$

or, $\text{Rs } 9,750 = 0.015x$

∴ $x = \text{Rs } 6,50,000$

Hence, the total sales of the next month should be Rs 6,50,000 so that he/she can receive a total income of Rs 31,350 in the month.

3. ***Mr. Bibek is an online salesperson in an online shopping store. His monthly salary Rs 18,700 and 2% commission is given to him when the monthly sales is more than 5 lakh rupees. If the sales of the store in a month is Rs 7,20,000, calculate his total income of the month.***

Solution:

Here, the monthly salary of a salesperson = Rs 18,700

The fixed sales of the month = 5 lakh rupees = Rs 5,00,000

The total sales of the month = Rs 7,20,000

The sales eligible for commission = Rs 7,20,000 – Rs 5,00,000 = Rs 2,20,000

Now, the commission received by the sales person = 2% of Rs 2,20,000 = Rs 4,400

Hence, the income of the sales person in the month = salary + commission

$$= \text{Rs } 18,700 + \text{Rs } 4,400 = \text{Rs } 23,100$$

- 4. Mrs. Nepali draws Rs 19,800 as her monthly salary in a wholesale cosmetic shop and a certain commission is given as per the monthly sales. If the sales of the month is Rs 12,00,000 and her total income of the month including commission is Rs 31,800, find the rate of commission.**

Solution:

Here, monthly salary = Rs 19,800 and the income of the month = Rs 31,800

∴ Commission amount received in the next month = Rs 31,800 – Rs 19,800 = Rs 12,000

The total sale of the month = Rs 12,00,000

Now, commission percent = $\frac{\text{Commission amount}}{\text{Total sales}} \times 100\% = \frac{\text{Rs } 12,000}{\text{Rs } 12,00,000} \times 100\% = 1\%$

Hence, the rate of the commission is 1%.

Extra Questions

- Mr. Shrestha is a salesperson in a hardware shop. His monthly salary Rs 22,500 and 1.5% commission is given to him when the monthly sales is more than 10 lakh rupees. If the sales of the hardware shop in a month is Rs 17,50,000, calculate his total income of the month. [Ans: Rs 33,750]
- Aravi draws Rs 15,000 as her monthly salary in a wholesale fancy shop and a certain commission is given as per the monthly sales. If the sales of the month is Rs 10,00,000 and her total income of the month including commission is Rs 27,500, find the rate of commission. [Ans: 1.25%]
- The monthly salary of Sanjay, a salesman of a departmental store, is Rs 25,000 and an additional incentive of 1.5% on the total monthly sale is provided as commission.
 - Find his total income in a month if he makes a total sale of Rs 5,55,000 in the month.
 - What should be his total sale in the next month so that he can receive a total income of Rs 38,320 in the month? [Ans: Rs 33,250; Rs 8,88,000]

C. Income Tax**Teaching Activities**

- Ask about the yearly income of parents of the student and discuss upon the tax is to be paid to the government.
- Divide the students into groups and give them to study the printed form of the present rates of income taxes fixed by Inland Revenue Department (IRD) and discuss upon the following questions
 - What is income tax?
 - Which authentic body is responsible to collect the tax?
 - Why should we pay tax to the government?
 - What do you mean by taxable income?
 - Which incomes are entitles for tax rebate?
 - What is the rate of social security tax?

3. Present the income tax slabs for individual and married couple separately on a chart paper.
4. Tell the students to write the important notes and formulae on the colourful chart paper as project work
5. With some related examples, let the students identify the following formulae then explain them one by one.
 - (i) Taxable income = Yearly income – tax free income
 - (ii) Income tax = rate of tax of taxable income
 - (iii) Tax rate = $\frac{\text{Income tax}}{\text{Taxable income}} \times 100\%$

Solution of selected problems from Vedanta Excel in Mathematics

1. ***Mr Sunil Jha is a Secondary level Mathematics Teacher in a community school. His monthly salary is Rs 38,700 and one month's salary as Dashain Bonus. 10% of his salary is deducted to deposit in his provident fund. If his marital status is single, calculate the annual income tax paid by him.***

Solution:

Here, his monthly income after deducting the provident fund

$$= \text{Rs } 38,700 - 10\% \text{ of Rs } 38,700 = \text{Rs } 34,830$$

His annual income with Dashain bonus = $12 \times \text{Rs } 34,830 + \text{Rs } 38,700 = \text{Rs } 4,56,660$

Social security tax up to Rs 4,00,000 = 1% of Rs 4,00,660 = Rs 4,000

Taxable income = Rs 4,56,660 – Rs 4,00,000 = Rs 56,660

Income tax for Rs 56,660 = 10% of Rs 56,660 = Rs 5,666

Hence, the total annual income tax paid by him = Rs 4,000 + Rs 5,666 = Rs 9666

2. ***After deducting 10% provident fund a married person draws Rs 40,500 salary per month and one month's salary as festival bonus. He/she pays Rs 14,500 annually as premium of his/her insurance. Calculate the annual income tax paid by the person.***

Solution:

Here, his monthly income after deducting the provident fund = Rs 40,500

Let the monthly income of the person be Rs x.

Then, $x - 10\% \text{ of } x = \text{Rs } 40,500 \therefore x = \text{Rs } 45,000$

His annual income with Dashain bonus = $12 \times \text{Rs } 40,500 + \text{Rs } 45,000 = \text{Rs } 5,31,000$

Taxable income after premium of insurance = Rs 5,31,000 – Rs 14,500 = Rs 5,16,500

Social security tax up to Rs 4,50,000 = 1% of Rs 4,50,660 = Rs 4,500

Taxable income for 10% tax = Rs 5,16,500 – Rs 4,50,000 = Rs 66,500

Income tax for Rs 66,500 = 10% of Rs 66,500 = Rs 6,650

Hence, the total annual income tax = Rs 4,500 + Rs 6,650 = Rs 11,150

3. ***Mr. Sayad Sharma an unmarried employee of a UN Project draws monthly salary of Rs 51,000 after deducting 10% salary in his provident fund and 5% in citizen investment trust. He also receives a Dashain Bonus of one month's salary. He pays Rs 22,000 annually as the premium of his life insurance. How much income tax does he pay in a year?***

Solution:

Here, his monthly income after deducting 10% provident fund and 5% citizen investment trust = Rs 51,000

Let his monthly income before deducting the provident fund and citizen investment trust be Rs x.

Then, $x - (10 + 5) \% \text{ of } x = \text{Rs } 40,500 \therefore x = \text{Rs } 60,000$

His annual income with Dashain bonus = $12 \times \text{Rs } 51,000 + \text{Rs } 60,000 = \text{Rs } 6,72,000$

Taxable income after premium of insurance = $\text{Rs } 6,72,000 - \text{Rs } 22,000 = \text{Rs } 6,50,000$

Social security tax up to $\text{Rs } 4,00,000 = 1\% \text{ of } \text{Rs } 4,00,660 = \text{Rs } 4,000$

Taxable income for 10% tax = $\text{Rs } 5,00,000 - \text{Rs } 4,00,000 = \text{Rs } 1,00,000$

Income tax for $\text{Rs } 1,00,000 = 10\% \text{ of } \text{Rs } 1,00,000 = \text{Rs } 10,000$

Taxable income for next 20% tax = $\text{Rs } 6,50,000 - \text{Rs } 5,00,000 = \text{Rs } 1,50,000$

Income tax for $\text{Rs } 1,50,000 = 20\% \text{ of } \text{Rs } 1,50,000 = \text{Rs } 30,000$

Hence, the total annual income tax = $\text{Rs } 4,000 + \text{Rs } 10,000 + \text{Rs } 30,000 = \text{Rs } 44,000$.

Extra Questions

1. The monthly income of an unmarried individual is $\text{Rs } 45,000$. If 1% social security tax is charged up to $\text{Rs } 4,00,000$. Then 10% and 20% are charged for the next $\text{Rs } 1,00,000$ and $\text{Rs } 2,00,000$ respectively. Calculate the annual income tax paid by the individual.
[Ans: $\text{Rs } 22,000$]
2. Rumakanta Jha is a married professor. His monthly income is $\text{Rs } 55,000$. If 1% social security tax is charged up to $\text{Rs } 4,50,000$; 10% tax for the income from $\text{Rs } 4,50,000$ to $\text{Rs } 5,50,000$ and 20% tax from $\text{Rs } 5,50,000$ to $\text{Rs } 7,50,000$ are to be paid. Calculate the annual income tax paid by the individual.
[Ans: $\text{Rs } 36,500$]
3. Mrs Pandey is a Branch Manager of a commercial bank. Here monthly salary is $\text{Rs } 95,400$ and 10% of her salary is deducted as provident fund. She pays $\text{Rs } 24,520$ as the premium of her life insurance. If 1% social security tax is levied upon the income of $\text{Rs } 4,50,000$, 10%, 20% and 30% taxes are levied upon the next incomes of $\text{Rs } 1,00,000$, $\text{Rs } 2,00,000$ and up to $\text{Rs } 12,50,000$ respectively, how much income tax should she pay in a year?
[Ans: $\text{Rs } 1,31,240$]

D. Dividend

Teaching Activities

1. Recall the bonus.
2. Create a short story about dividend and tell in the class.
3. Define dividend as the certain amount distributed among the shareholders of a corporation as per the number of shares from the net profit.
4. Ask the following questions during classroom discussion
 - (i) What is dividend?
 - (ii) Among whom the dividend is distributed?
 - (iii) Tell the difference between the bonus and the dividend.
5. Make a discussion upon the following formulae with examples.
 - (i) $\text{Dividend} = \text{Rate of dividend (in \%)} \times \text{Net profit}$
 - (ii) $\text{Dividend} = \text{Value of dividend per share} \times \text{Total number of shares}$
 - (iii) $\text{Value of dividend for each share} = \frac{\text{Dividend amount}}{\text{No. of shares}}$
 - (iv) $\text{Rate of dividend (in \%)} = \frac{\text{Dividend amount}}{\text{No. of shares}} \times 100\%$

Solution of selected problems from Vedanta Excel in Mathematics

1. **Mrs Rai bought 250 shares out of 10,000 shares from a financial company. The company earned a net profit of Rs 85,20,000 and declared 17% dividend to its shareholders. Calculate the amount of dividend received by Mrs Rai.**

Solution:

Here, dividend amount = 17% of Rs 85,20,000 = Rs 14,48,400

$$\text{Value of dividend for each share} = \frac{\text{Dividend amount}}{\text{No. of shares}} = \frac{\text{Rs } 14,48,400}{10,000} = \text{Rs } 144.84$$

$$\therefore \text{Dividend for 250 shares} = 250 \times \text{Rs } 144.84 = \text{Rs } 36,210$$

Hence, Mrs Rai received Rs 36,210 as dividend.

2. **A Cable Car Company sold 3000 shares to the local people. The company earned a profit of Rs 1,20,00,000 in a year and distributed a certain percent of profit as dividend. If a shareholder who has bought 125 shares received Rs 1,10,000 dividend, what percent of profit was distributed as dividend?**

Solution:

$$\text{Here, dividend amount distributed for 1 share} = \frac{\text{Rs } 1,10,000}{125} = \text{Rs } 880$$

$$\therefore \text{Dividend amount distributed for 3000 shares} = 3000 \times \text{Rs } 880 = \text{Rs } 26,40,000$$

$$\text{Net profit} = \text{Rs } 1,20,00,000$$

$$\text{Now, rate of dividend} = \frac{\text{Dividend amount}}{\text{Net profit}} \times 100\% = \frac{\text{Rs } 26,40,000}{\text{Rs } 1,20,00,000} \times 100\% = 22\%$$

Hence, 22 of the profit was distributed as dividend.

3. **Mr. Dhurmus bought 500 shares out of 10,000 shares sold by a commercial bank. The bank earned some profit and distributed 14% of the net profit as the dividend in a year. If Dhurmus received Rs 1,03,600 in the year, find the net profit of the bank.**

Solution:

$$\text{Here, dividend received for 500 shares} = \text{Rs } 1,03,600$$

$$\text{or, dividend received for 1 share} = \text{Rs } \frac{1,03,600}{500} = \text{Rs } 207.20$$

$$\therefore \text{Dividend distributed for 10,000 shares} = 10,000 \times \text{Rs } 207.20 = \text{Rs } 20,72,000$$

Let the net profit of a year of the commercial bank be x.

$$\text{Then, dividend amount} = \text{Rate of dividend} \times \text{Net profit}$$

$$\text{or, Rs } 20,72,000 = 14\% \text{ of } x \quad \therefore x = \text{Rs } 1,48,00,000$$

Hence, the net profit of the bank was Rs 1,48,00,000

Extra Questions

- Mr Dahal bought 400 shares out of 20,000 shares from a Business Company. The company earned a net profit of Rs 2,25,00,000 and declared 15% dividend to its shareholders. Calculate the amount of dividend received by Mr Dahal. [Ans: Rs67,500]
- Mr Jeevan bought 225 shares out of 10,000 shares from commercial bank. If the company earned a profit of Rs 1,50,00,000 in a year and distributed a certain percent of profit as dividend. If Mr Jeevan received Rs 54,000 dividend, what percent of profit was distributed as dividend? [Ans:16%]
- Rajesh bought 300 shares out of 5,000 shares sold by an insurance company. The company earned some profit and distributed 20% of the net profit as the dividend in a year. If Rajesh received Rs 66,600 in the year, find the net profit of the bank.

[Ans:Rs 55,50,000]